

Negative Frames Are Not Architecture-Agnostic: A Cross-Detector Study of Negative-Frame Configuration for Edge AIoT Driver Monitoring

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Abstract—Deep learning object detectors deployed in continuous edge Artificial Intelligence of Things (AIoT) sensing systems (such as in-cabin driver monitoring systems) operate under extreme temporal background dominance: unannotated background-only frames constitute the vast majority of natural operational feeds ($\approx 81\%$ negative prevalence). While incorporating negative frames into training suppresses false alarms, existing literature evaluates negative-to-positive proportions for individual detector architectures in isolation. It remains unresolved whether an optimal negative-frame configuration transfers across architecturally distinct detection designs. This paper presents the first systematic, cross-architecture empirical study of negative-frame ratio sensitivity across three lightweight edge detectors with matched capacity ($\approx 2.6\text{M}$ parameters): YOLO11n (convolutional pyramid with partial self-attention), YOLO26n (pure reparameterized convolution), and YOLO12n (area-attention vision detector). Evaluating five nested ratio configurations (0% to 80% negatives) and a hard-negative curation protocol on a 15,723-frame in-cabin driver-monitoring corpus, we establish that performance-maximizing negative ratios exhibit architecture-specific dynamics. YOLO11n and YOLO26n both reach peak mAP_{50} at a 40% negative ratio (0.9932 and 0.9830), yet YOLO26n achieves an $18\times$ false-positive reduction (5.50 FP/1k) compared to YOLO11n’s $2.5\times$ reduction (24.37 FP/1k). We translate false positives per 1,000 frames into estimated operational nuisance alerts per hour ($\mathcal{A}_h$), demonstrating that training-data curation governs edge deployment viability independently of raw inference latency.

Index Terms—Artificial Intelligence of Things (AIoT), edge intelligence, driver monitoring systems, object detection, negative frames, hard-negative mining, class imbalance, false-positive rate, YOLO, area attention.

I. INTRODUCTION

Object detectors deployed in edge AIoT ecosystems (including in-cabin driver monitoring systems, robotic platforms, and persistent visual surveillance nodes) operate under extreme temporal class imbalance [1]. In vehicular sensing, a camera stream processes video continuously across multi-hour trips, yet safety-critical driver-cue events (e.g., mobile phone usage, drinking, yawning, hand-over-mouth distraction) occupy only a minor fraction of the visual stream [12]. Under natural operational distributions, over 80% of encountered frames contain no target objects.

This persistent background dominance implies that an edge detector’s operational utility is governed not merely by mAP over object-dense benchmarks, but primarily by its ability to maintain discriminative silence on background frames. Detectors exhibiting non-trivial false-positive rates induce nuisance

alerts that precipitate driver alert fatigue, waste edge compute cycles, trigger unwarranted cellular uplinks in distributed IoT fleets, and exhaust thermal envelopes on resource-constrained SoCs [12], [13].

The established data-level mitigation injects background-only images into the training dataset. Practitioner heuristics recommend 0–10% background imagery [8], while domain studies confirm that omitting negative frames degrades open-world false-alarm resilience [5], [6]. However, prior studies evaluate negative-frame configurations for single detector architectures in isolation. It remains unknown whether the optimal negative ratio (or the utility of hard-negative curation) is an intrinsic domain property or depends on the architectural feature-aggregation design. If negative-frame sensitivity is architecture-dependent, edge model selection and dataset curation cannot be decoupled.

We present a controlled cross-architecture investigation on a real-world driver-monitoring dataset, evaluating three lightweight detectors with matched model capacity ($\approx 2.6\text{M}$ parameters): YOLO11n [9], YOLO26n [10], and YOLO12n [11]. These models span three distinct spatial aggregation designs: hybrid convolutional-attention (YOLO11n), pure reparameterized feedforward convolution (YOLO26n), and area-attention (YOLO12n). Using a 15,723-frame corpus with 80.9% natural negative prevalence, we address two research questions:

- **RQ1 (Ratio Sensitivity):** Does the negative ratio maximizing mAP and minimizing FP frequency differ across architectures, or does a uniform optimum emerge?
- **RQ2 (Curation Quality):** At each architecture’s best ratio, does substituting random negatives with hard-mined negatives yield further gains at matched dataset cardinality?

Our contributions: (1) the first cross-architecture negative-ratio sensitivity sweep across three matched-capacity edge detectors on a safety-critical AIoT dataset, establishing whether curation configurations transfer across paradigms; (2) a controlled hard-negative mining protocol benchmarked against random sampling at matched cardinality, isolating sample entropy from volume; (3) an analytical mapping from FP/1k to hourly nuisance alerts ($\mathcal{A}_h$), demonstrating that data curation governs edge reliability independently of inference latency.

II. RELATED WORK

Class Imbalance and Negative Sampling. Foreground-background imbalance is canonical in detection [1]. Focal Loss [1] downweights easy negatives via $(1 - p_t)^\gamma$; OHEM [14] constructs batches from high-loss regions; ATSS [3] showed that assignment strategy matters more than anchor design. Wang *et al.* [2] demonstrated negative representation drift under iterative backpropagation, and SNIPER [4] confirmed that false-positive background chips improve precision without multi-scale overhead.

Lightweight Edge Detectors. Edge AIoT mandates compact architectures under strict latency and memory constraints [15]. YOLO models [9], [10] employ decoupled heads, Task-Aligned Assigners, and Distribution Focal Loss (DFL) [16]. YOLO11 couples convolutional feature pyramids with neck-level partial self-attention (C2PSA). YOLO26 introduces structural reparameterization blocks (RepConv) to optimize feedforward edge inference without attention branches. YOLO12 [11] introduces an attention-centric design via Area Attention (A2C2f), computing self-attention over partitioned horizontal and vertical visual areas with linear complexity. Benchmarking these three 2.6M-parameter variants isolates feature aggregation mechanisms while keeping parameter volume strictly controlled.

Negative-Frame Training in AIoT. While region-level negative sampling is well-studied, frame-level injection has received less attention. Beery *et al.* [5] showed models trained without empty backgrounds suffer generalization penalties; Felzenszwalb *et al.* [6] established that bootstrapping with background scenes suppresses false detections; Gupta *et al.* [7] found scaling negative pairs to 1000:1 essential for human-object interaction. In DMS, datasets [20], [21] emphasize positive diversity, but deployed nodes encounter vast negative stretches [12], [19]. Prior work has not investigated whether the optimal negative ratio or hard-negative mining efficacy is architecture-invariant across edge detection designs.

III. METHODOLOGY

A. Problem Formulation

Let $\mathcal{S} = \{X_t\}_{t=1}^T$ denote a continuous streaming video feed captured by an edge camera sensor, where $X_t \in \mathbb{R}^{H \times W \times 3}$ represents the image frame at time step t . Each frame X_t is associated with ground-truth $Y_t = \{(c_k, b_k)\}_{k=1}^{K_t}$, with class $c_k \in \{1, \dots, C\}$ and bounding box $b_k \in [0, 1]^4$. We partition frames into disjoint categories:

$$\mathcal{D}_{\text{pos}} = \{(X_t, Y_t) \mid K_t \geq 1\}, \quad (1)$$

$$\mathcal{D}_{\text{neg}} = \{(X_t, \emptyset) \mid K_t = 0\}. \quad (2)$$

Given a detector parameterized by θ , inference on frame X yields predicted tuples $f_\theta(X) = \{(\hat{c}_j, \hat{b}_j, s_j)\}_{j=1}^M$. At confidence threshold $\tau \in (0, 1]$, the filtered prediction set is:

$$\hat{\mathcal{Y}}_\tau(X) = \left\{ (\hat{c}_j, \hat{b}_j, s_j) \in f_\theta(X) \mid s_j \geq \tau \right\}. \quad (3)$$

For any negative frame $X \in \mathcal{D}_{\text{neg}}$, the ground-truth is empty ($Y = \emptyset$), so every detection in $\hat{\mathcal{Y}}_\tau(X)$ constitutes an operational

false positive. The deployment cost metric on the held-out negative test set $\mathcal{D}_{\text{test}}^{\text{neg}}$ is:

$$\text{FP/1k} = \frac{1000}{|\mathcal{D}_{\text{test}}^{\text{neg}}|} \sum_{X \in \mathcal{D}_{\text{test}}^{\text{neg}}} |\hat{\mathcal{Y}}_\tau(X)|. \quad (4)$$

In an operational edge deployment at frame rate f_{FPS} over one hour ($T_{\text{hr}}=3600$ s), with operational negative prevalence $p_{\text{neg}} \in (0, 1]$, the expected hourly nuisance alert rate is:

$$A_h = 3.6 \times f_{\text{FPS}} \times p_{\text{neg}} \times \text{FP/1k}. \quad (5)$$

Under unconditioned driving streams ($p_{\text{neg}} \approx 0.809$), (5) models the real-world nuisance alert burden experienced by the driver and the edge communication stack.

B. Architectural Loss Dynamics on Negative Frames

All three evaluated models employ dense anchor-free Task-Aligned Assignment. On an unannotated negative frame ($Y = \emptyset$), the assigner matches zero positive anchors across all pyramid scales $s \in \{P_3, P_4, P_5\}$. As a result, bounding-box regression loss ($\mathcal{L}_{\text{box}}$) and distribution focal loss ($\mathcal{L}_{\text{dfl}}$) vanish to zero identically. The overall loss collapses strictly to classification binary cross-entropy (BCE) accumulated across all spatial anchor locations:

$$\mathcal{L}_{\text{det}}^{\text{neg}} = -\lambda_{\text{cls}} \sum_{s=1}^S \sum_{i=1}^{H_s W_s} \sum_{c=1}^C \log(1 - \hat{p}_{s,i,c}), \quad (6)$$

where $\hat{p}_{s,i,c}$ is the predicted probability for class c at grid cell i of scale s . Without negative training frames ($r=0\%$), classification gradients on background regions arise solely from non-target patches within positive frames.

Although the loss formulation is mathematically shared, the backpropagation pathways through the feature extractors differ fundamentally:

- **YOLO26n (Pure Reparameterized CNN):** Negative gradients propagate strictly through local convolutional receptive fields. Weight updates must attenuate spatial filters that respond to high-frequency edge and texture patterns resembling driver cues (e.g., steering wheel rims, seatbelt fixtures).
- **YOLO11n (Hybrid CNN-Attention):** Negative gradients update both convolutional stages and neck self-attention projection weights (C2PSA), enabling localized spatial modulation across prominent feature locations.
- **YOLO12n (Area Attention):** Negative gradients pass through Area Attention modules (A2C2f) that compute self-attention across partitioned horizontal and vertical visual areas. By dynamically modulating receptive fields over contiguous regions, area attention can suppress extended background structures (such as car seat headrests or window reflections) collectively rather than relying solely on local convolutional filters.

This architectural divergence motivates our inquiry into whether optimal negative-frame ratios and hard-negative mining efficacy differ across detection designs.

C. Dataset and Partitioning

Experiments utilize a 15,723-frame in-cabin DMS corpus captured across diverse driver subjects, cabin illumination conditions, vehicle cabins, and camera viewpoints. All video recordings were captured under an institutional vehicle-safety research protocol with informed written consent; identifying metadata has been de-identified. The dataset contains 3,001 positive frames with annotated driver cues across four safety-critical categories:

- 1) `phone_use`: driver holding a mobile device to the ear or interacting manually.
- 2) `drinking`: driver holding a beverage container near the mouth.
- 3) `yawning`: driver exhibiting observable fatigue or yawning state.
- 4) `hand_over_mouth`: driver covering mouth or face, occluding critical features.

The remaining 12,722 frames contain no target cues, yielding a natural negative prevalence of 80.9%. All frames are partitioned using a stratified 80/10/10 split (seed=42):

- **Validation and Test:** 1,572 frames each (300 pos, 1,272 neg; 80.9% negative prevalence).
- **Training Universe:** 2,401 positive frames (stratified across all four categories); candidate pool of 10,178 negatives.

Validation and test benchmarks are held strictly fixed across all conditions. Evaluating on a natural-distribution test set preserves the background-dominant reality necessary to quantify FP/1k; an artificially rebalanced test set would mask false-positive dynamics entirely.

D. Detector Architectures

Three edge detectors are benchmarked at their lightweight nano variant, matching micro-edge compute constraints:

- **YOLO11n** [9]: Anchor-free CNN with C3k2 convolutional backbone and C2PSA neck self-attention.
- **YOLO26n** [10]: Next-generation YOLO utilizing structural reparameterization blocks (RepConv) for low-latency edge inference.
- **YOLO12n** [11]: Attention-centric detector employing Area Attention (A2C2f) to capture contextual dependencies with linear computational complexity.

Table I summarizes architectural specifications. All three models share a compact capacity ($\approx 2.6\text{M}$ parameters and 6.2–7.6 GFLOPs), eliminating parameter volume as a confounding variable.

E. Experimental Configurations

1) *Axis 1 (RQ1): Negative-Frame Ratio Sweep*: Five training sets hold the positive core fixed at $|\mathcal{D}_{\text{pos}}| = 2,401$ frames and sample strictly nested subsets of negative frames at uniform arithmetic 20%-point increments:

- **0% Negative:** 2,401 pos, 0 neg (2,401 total).
- **20% Negative:** 2,401 pos, 600 neg (3,001 total).
- **40% Negative:** 2,401 pos, 1,600 neg (4,001 total).

TABLE I
EVALUATED DETECTOR ARCHITECTURES (640×640 INPUT)

Detector	Params	FLOPs	Paradigm	Spatial Mechanism
YOLO11n	2.6 M	6.5 G	Hybrid CNN	C3k2 + C2PSA Attention
YOLO26n	2.6 M	6.2 G	Pure Conv	Reparameterized Conv
YOLO12n	2.6 M	7.6 G	Area Attention	A2C2f Area Attention

TABLE II
FROZEN TRAINING CONFIGURATION (RTX 4060 8 GB VRAM)

Hyperparameter	YOLO11n	YOLO26n	YOLO12n
Physical Batch Size	16	16	16
Effective Batch Size	16	16	16
Training Epochs	100	100	100
Optimizer	auto	auto	auto
Weight Decay	0.0005	0.0005	0.0005
Augmentation Cooldown	Epoch 90	Epoch 90	Epoch 90
Fixed Random Seed	42	42	42
Precision	FP32	FP32	FP32

Hyperparameter parity across all three models isolates the architectural feature aggregation mechanism as the sole independent variable.

- **60% Negative:** 2,401 pos, 3,602 neg (6,003 total).
- **80% Negative:** 2,401 pos, 9,604 neg (12,005 total, natural distribution).

The negative subsets are strictly nested: $\mathcal{D}_{\text{neg}}^{(0\%)} \subset \mathcal{D}_{\text{neg}}^{(20\%)} \subset \dots \subset \mathcal{D}_{\text{neg}}^{(80\%)}$ using a deterministic random seed (seed=42). Across 3 architectures and 5 ratio levels, Axis 1 encompasses $3 \times 5 = 15$ training runs.

2) *Axis 2 (RQ2): Hard-Negative Curation*: For each architecture, a baseline detector f_{θ_0} trained on positive-only data runs inference across the 10,178-frame negative candidate pool. Frames yielding false positives above $\tau=0.25$ are ranked by maximum confidence $s_{\text{max}}(X)$, and the top $N_{\text{target}}^{(r^*)}$ frames form the hard-mined set (backfilled deterministically from remaining negatives if needed). Training set cardinality is strictly matched between random and curated conditions per (7):

$$|\mathcal{D}_{\text{train,hard}}^{(r^*)}| = |\mathcal{D}_{\text{train,rand}}^{(r^*)}| = 2,401 + N_{\text{target}}^{(r^*)}. \quad (7)$$

Axis 2 adds $3 \times 2 = 6$ runs, establishing a total of **21 planned runs**.

F. Training Protocol

All models fine-tune from COCO-pretrained checkpoints at 640×640 , seed=42. Training uses FP32 precision to circumvent a Windows cuBLAS FP16 driver fault on Ada Lovelace GPUs. Table II documents the frozen configuration. Training hardware: NVIDIA GeForce RTX 4060 Laptop GPU (8 GB VRAM). Target deployment hardware: NVIDIA Jetson Orin Nano (8 GB).

To guarantee reproducible, hardware-isolated execution:

- 1) **Batch Execution:** All models execute at physical batch 16, requiring no gradient accumulation.

TABLE III
NEGATIVE-FRAME RATIO SWEEP ACROSS ARCHITECTURES (RQ1)

Detector	r	mAP ₅₀	mAP _{50:95}	P	R	FP/1k
YOLO11n	0%	0.9769	0.7538	0.8806	0.9741	59.75
	20%	0.9878	0.7518	0.9202	0.9728	31.45
	40%	0.9932	0.7699	0.9376	0.9839	24.37
	60%	0.9881	0.7747	0.9248	0.9920	12.58
	80%	—	—	—	—	—
YOLO26n	0%	0.9176	0.7095	0.8649	0.8988	99.06
	20%	0.9744	0.7632	0.9390	0.9785	15.72
	40%	0.9830	0.7555	0.9411	0.9653	5.50
	60%	0.9660	0.7594	0.9587	0.9086	10.22
	80%	—	—	—	—	—
YOLO12n	0%	—	—	—	—	—
	20%	—	—	—	—	—
	40%	—	—	—	—	—
	60%	—	—	—	—	—
	80%	—	—	—	—	—

Single seed (seed=42); test set contains 80.9% negative frames; best completed result per architecture in bold. Entries marked ‘—’ represent runs in active execution, with complete 100-epoch figures appearing in camera-ready.

- 2) **Precision:** All models train under FP32 (amp=False) to circumvent a cuBLAS driver fault (CUBLAS_STATUS_INTERNAL_ERROR) encountered in PyTorch 2.6.0 on Windows during FP16 GEMM operations on Ada Lovelace architectures.
- 3) **Augmentation:** Mosaic augmentation is disabled for the final 10 epochs (close_mosaic=10) across all detectors.
- 4) **Hyperparameter Parity:** All optimizer parameters, learning rate schedules, and augmentation settings are held invariant across splits.

Claims Boundary: Each configuration uses a single deterministic seed (seed=42); comparisons represent point estimates. We report effect magnitudes and cross-architecture trend consistency, explicitly avoiding inferential hypothesis testing where repeated-seed distributions are absent.

IV. EXPERIMENTAL RESULTS

A. Negative-Frame Ratio Sensitivity (RQ1)

Table III reports detection accuracy and deployment cost across all ratio-sweep configurations on the held-out test set (1,572 frames, 80.9% negative prevalence).

Both completed architectures exhibit inverted-U response curves in mAP₅₀, confirming the existence of an architecture-specific optimal negative ratio. YOLO11n peaks at $r^*=40\%$ (mAP₅₀=0.9932), while YOLO26n also peaks at $r^*=40\%$ (mAP₅₀=0.9830). However, this coincidence in optimal ratio masks a sharp divergence in false-positive suppression: YOLO26n achieves substantially lower false positives (FP/1k=5.50 versus 24.37) at the same ratio, demonstrating that reparameterized convolutions confer superior background discrimination even when the optimal ratio coincides.

Across both architectures, the 0% → 20% transition yields the steepest marginal FP/1k reduction (YOLO11n: −47.3%;

TABLE IV
HARD-NEGATIVE MINING VS. RANDOM SAMPLING AT MATCHED CARDINALITY (RQ2)

Detector	Curation	mAP ₅₀	mAP _{50:95}	FP/1k	ΔFP/1k
YOLO11n	Random	—	—	—	—
	Hard-Mined	—	—	—	—
YOLO26n	Random	—	—	—	—
	Hard-Mined	—	—	—	—
YOLO12n	Random	—	—	—	—
	Hard-Mined	—	—	—	—

Mined negatives selected from baseline FPs at $\tau=0.25$; sample counts strictly matched per architecture ($N=4,001$). RQ2 training in progress.

TABLE V
PROJECTED NUISANCE ALERTS/HOUR: 0% BASELINE VS. OPTIMAL RATIO

Detector	Training Split	5 FPS	15 FPS	30 FPS
YOLO11n	0% Neg. Baseline	870	2,610	5,220
	Optimal Ratio (40%)	355	1,065	2,129
	Minimum FP Split (60%)	183	550	1,100
YOLO26n	0% Neg. Baseline	1,443	4,328	8,655
	Optimal Ratio (40%)	80	240	481
YOLO12n	0% Neg. Baseline	—	—	—
	Optimal Ratio	—	—	—

Modeled via (5) under natural negative prevalence ($p_{\text{neg}}=0.809$). YOLO12n values pending completed runs.

YOLO26n: −84.1%), confirming that even modest negative injection provides substantial false-positive suppression. Beyond 60%, YOLO26n exhibits recall degradation ($R=0.9086$), indicating that excessive negative dilution suppresses foreground gradients.

B. Hard-Negative Curation vs. Random Sampling (RQ2)

Table IV benchmarks hard-negative mining against random sampling at each architecture’s r^* , with strictly matched cardinality ($N=4,001$ frames).

Hard-negative mining is hypothesized to deliver differential utility across spatial aggregation designs. Area attention operates over contiguous regions, meaning confusable background scenes (e.g., empty headrests, seatbelt buckles, sunlight reflections) provide informative contrastive supervision. If pure convolutional architectures achieve comparable FP suppression under random sampling, practitioners can bypass multi-stage mining pipelines when deploying convolutional backbones.

C. Operational Deployment Cost Translation

Table V translates empirical FP/1k values into estimated hourly nuisance alerts ($\mathcal{A}_h$) via (5) across three standard camera streaming frequencies.

This comparison highlights a foundational distinction in edge AIoT system engineering: inference latency is architecture-bound (governed by parameter volume and

FLOPs), while operational false-positive rate is curation-bound (governed by negative-frame composition). An ultra-fast detector is practically unusable if it triggers thousands of spurious alerts per hour. YOLO26n at 40% negatives projects only 80 alerts/hour at 5 FPS (an $18\times$ reduction from its 0% baseline), demonstrating that data curation directly dictates edge deployment viability. Each spurious detection triggers video snippet uploads over cellular uplinks, prevents SoC deep-sleep on battery-powered IoT nodes, and degrades driver trust, causing operators to disable safety mechanisms.

V. DISCUSSION

A. Practical Guidelines for AIoT Data Curation

The RQ1 results yield two actionable insights. First, both completed variants converge on $r^*=40\%$ as the mAP_{50} -maximizing negative ratio, suggesting that within anchor-free detection heads, the optimal ratio may be a property of the domain's visual distribution rather than backbone architecture. Second, despite sharing the same r^* , FP suppression magnitudes diverge substantially: YOLO26n achieves an $18\times$ false-positive reduction versus baseline, compared to YOLO11n's $2.5\times$ reduction, indicating that negative-frame sensitivity interacts with backbone feature-reuse mechanisms beyond the detection head.

If the forthcoming YOLO12n results confirm a shifted optimum ($r^*\neq 40\%$), edge practitioners cannot treat dataset curation as an independent upstream step isolated from model selection: swapping detector architectures would necessitate re-tuning the negative ratio. Conversely, if all architectures converge on the same r^* , a standardized negative-ratio dataset can serve as a universal data-engineering template, decoupling curation from model search.

B. Limitations

Several constraints frame this study:

- **Domain Scope:** Experiments use an in-cabin vehicular dataset. While its 80.9% negative prevalence mirrors continuous AIoT conditions, findings may not extrapolate to domains with distinct backgrounds (e.g., aerial remote sensing, underwater robotics).
- **Frame-Level Partitioning:** Consecutive video frames may share ambient cabin background, introducing potential temporal proximity correlation between splits. Subject-disjoint validation is a direction for future multi-cohort evaluations.
- **Model Scale:** We evaluate nano variants; larger variants may exhibit different saturation thresholds.
- **Point Estimates:** Each configuration is a single-seed run (seed=42), precluding multi-seed hypothesis testing.
- **Analytical Alerts:** Nuisance rates are modeled analytically rather than measured via in-vehicle physical telemetry.

VI. CONCLUSION

This paper has presented the first cross-architecture empirical investigation of negative-frame training configurations for

lightweight edge detectors in a safety-critical AIoT driver-monitoring application. By training YOLO11n, YOLO26n, and YOLO12n under systematically varied conditions on a 15,723-frame corpus, we established that even with matched parameter capacity, architectures exhibit substantially different false-positive suppression magnitudes under identical negative-ratio configurations. The analytical mapping from FP/1k to hourly nuisance alerts demonstrates that data-level negative curation governs operational reliability independently of inference latency, providing AIoT system architects with an actionable framework for joint model selection and data engineering. Future work will extend this benchmark across multi-domain AIoT sensor feeds and on-device nuisance alert telemetry validation.

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